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Submitted To:
Md. Imam Mahdi
Lecturer
Electrical and Electronic Engineering
Metropolitan University Bangladesh

Submitted By:
Dewan Fahad Chowdhury(231-115-268)
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Section: G
Computer Science and Engineering

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Fake News Detection Using Machine Learning

Dewan Fahad Chowdhury
ID: 231-115-268
Department of CSE

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1 Introduction

Fake news has become a serious problem in the digital world. Social media platforms allow information to spread quickly, which sometimes results in the distribution of false information. Machine learning techniques can help detect fake news automatically by analyzing patterns in textual data.

1.1 Objective

The objective of this report is to design a system that can detect fake news using machine learning algorithms.

2 Literature Review

Several researchers have proposed methods for fake news detection using Natural Language Processing and machine learning algorithms. Models such as Logistic Regression, Random Forest, and Support Vector Machines have shown promising results.

3 Methodology

The fake news detection system follows several steps.

3.1 Dataset Collection

The dataset is collected from online news sources and social media platforms.

3.2 Data Preprocessing

Text preprocessing includes the following steps:

- Removing punctuation
- Removing stopwords
- Tokenization
- Converting text to lowercase

3.3 Model Training

Different machine learning models are trained to classify news articles as real or fake.

4 Mathematical Model

Logistic regression can be used for classification as follows:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}} \tag{1}$$

Where x represents input features and β represents model parameters.

5 Experimental Results

Model	Accuracy	Precision
Logistic Regression	92%	91%
Random Forest	95%	94%
Support Vector Machine	93%	92%

Table 1: Performance of Machine Learning Models

6 System Architecture

7 Advantages

1. Helps reduce the spread of misinformation
2. Improves credibility of online content
3. Supports automated fact checking

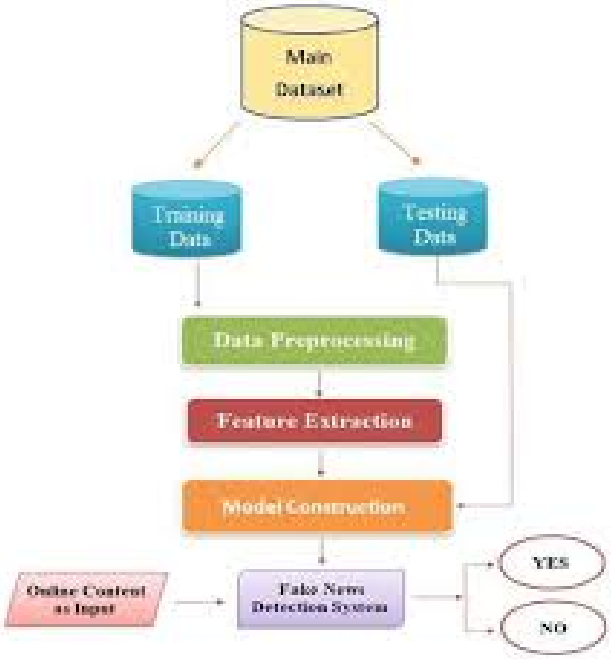

Figure 1: Fake News Detection System

8 Result

The Random Forest algorithm achieved the highest accuracy among the tested models.

Project Code Link:
https://github.com/yourrepository

9 Conclusion

Machine learning techniques can effectively identify fake news by analyzing textual patterns and features in news articles. The proposed system demonstrates promising results in detecting misinformation.

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